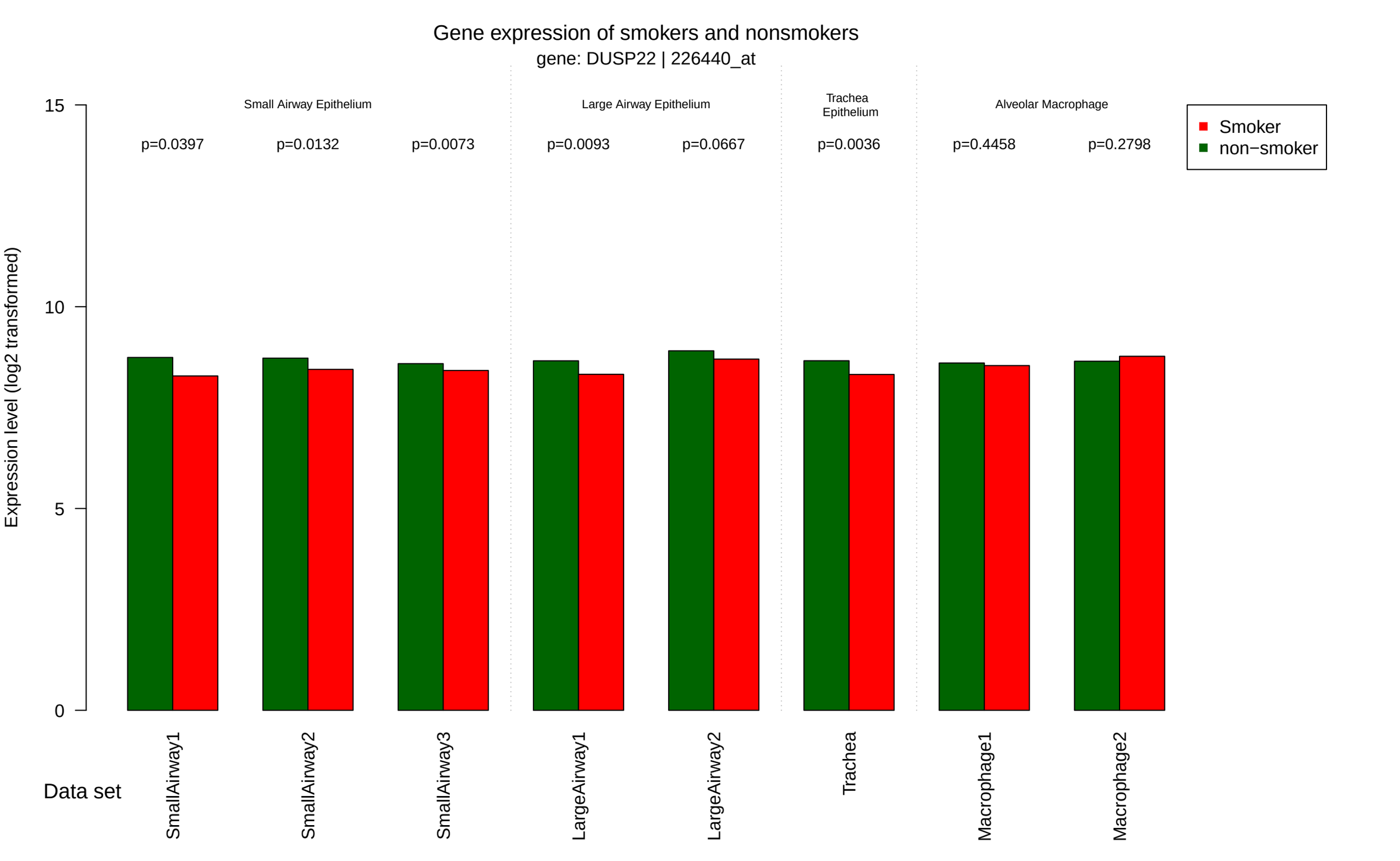
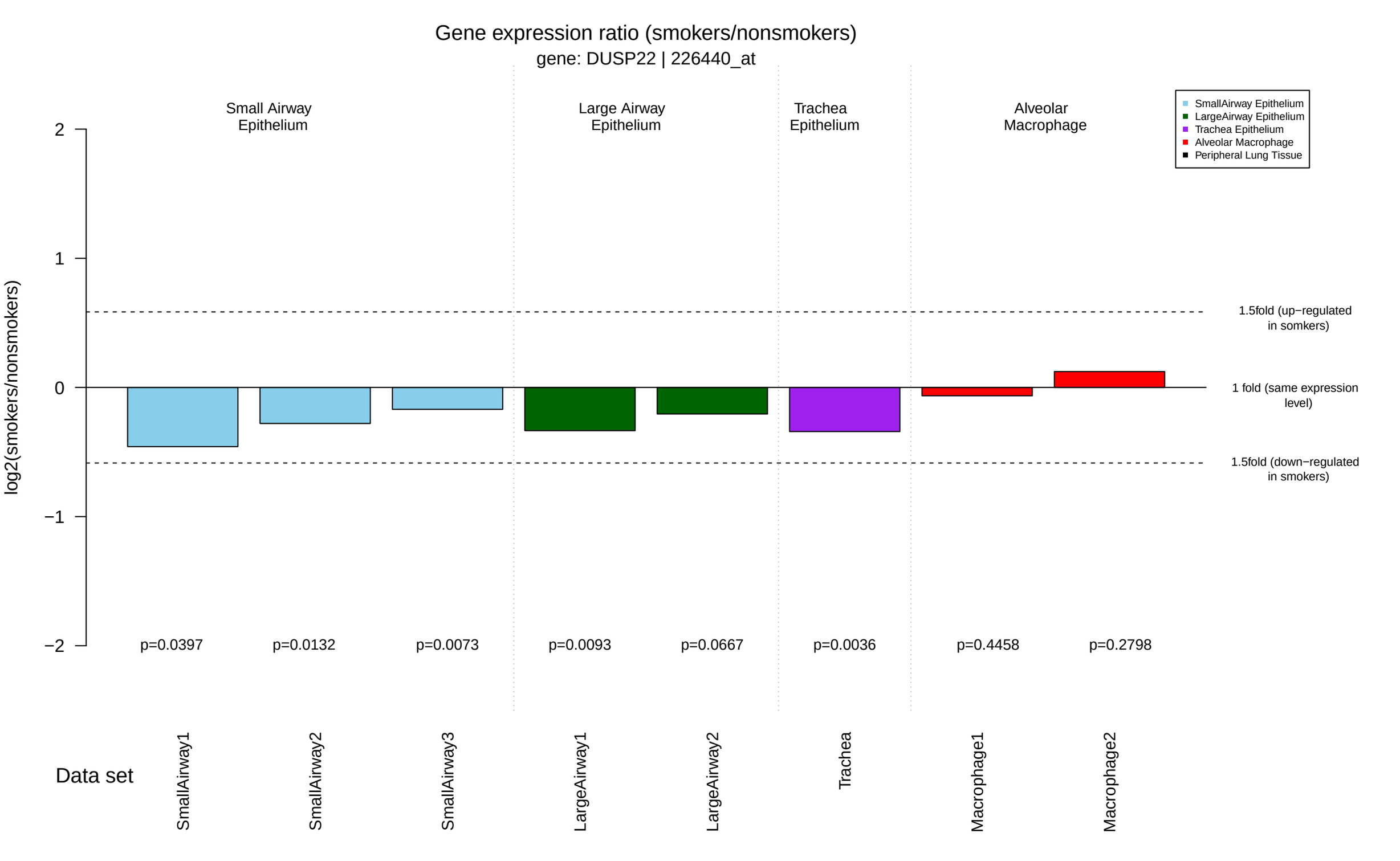
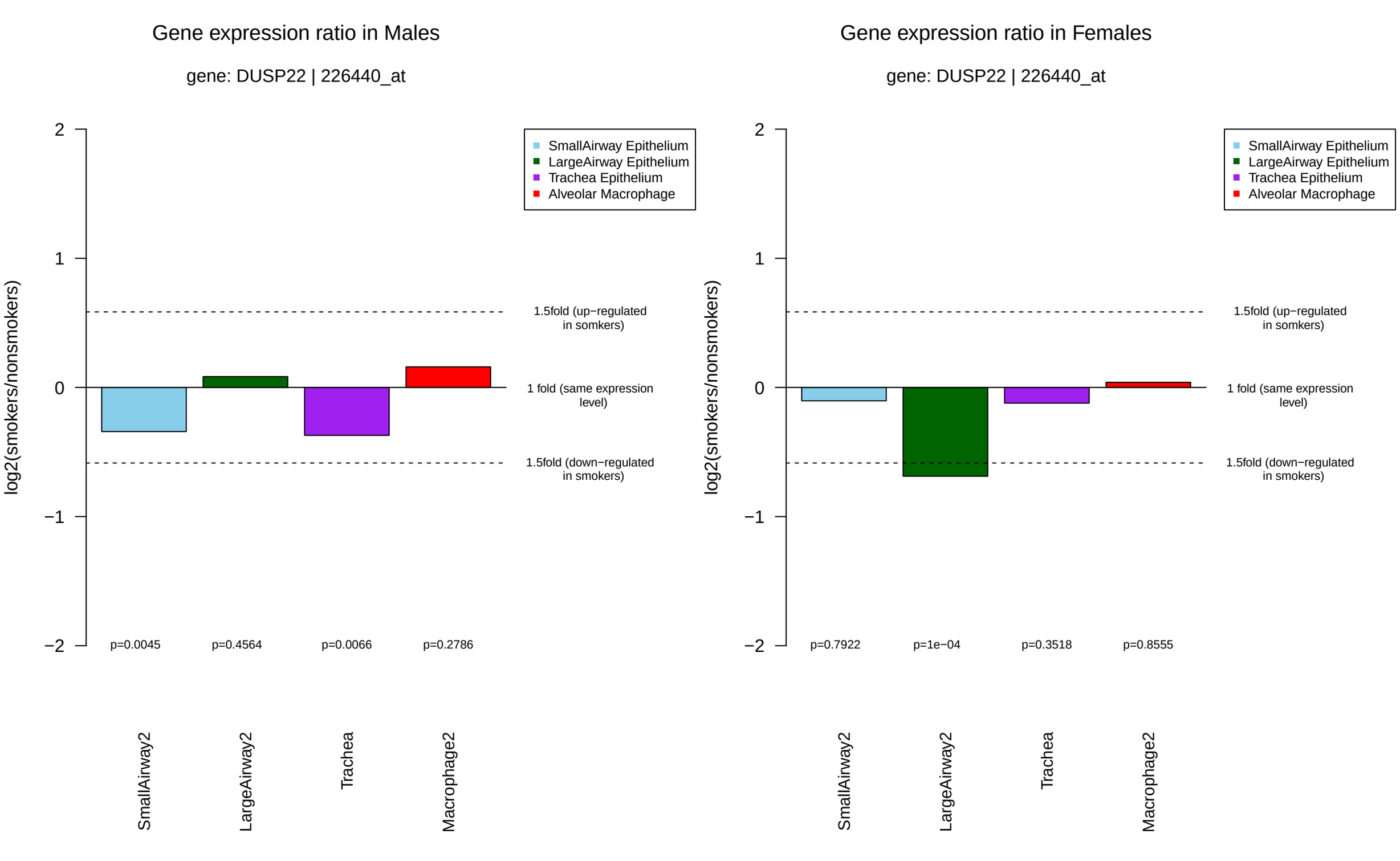
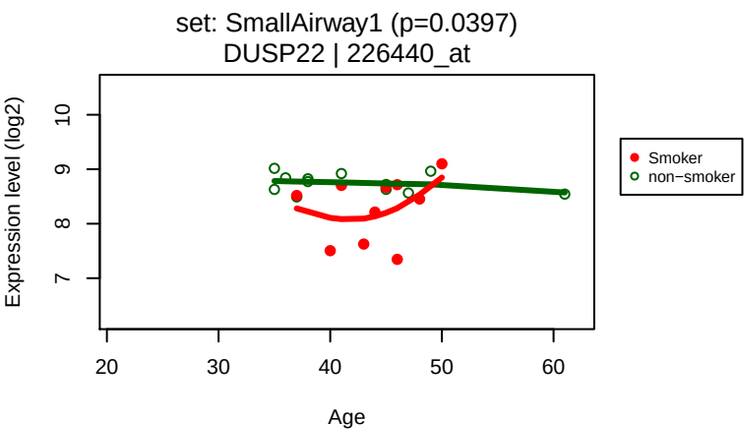
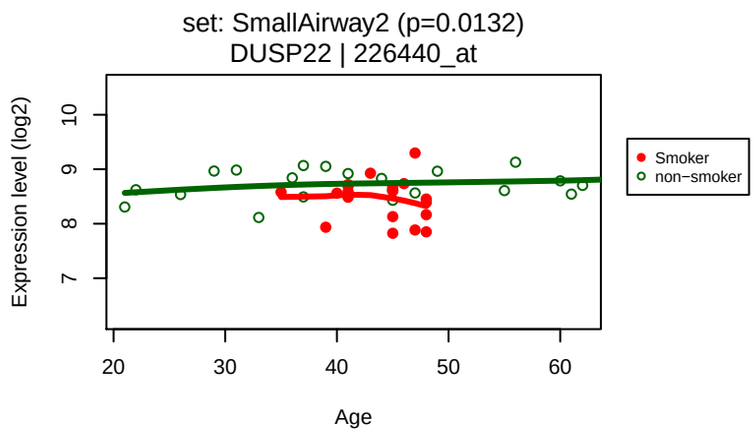
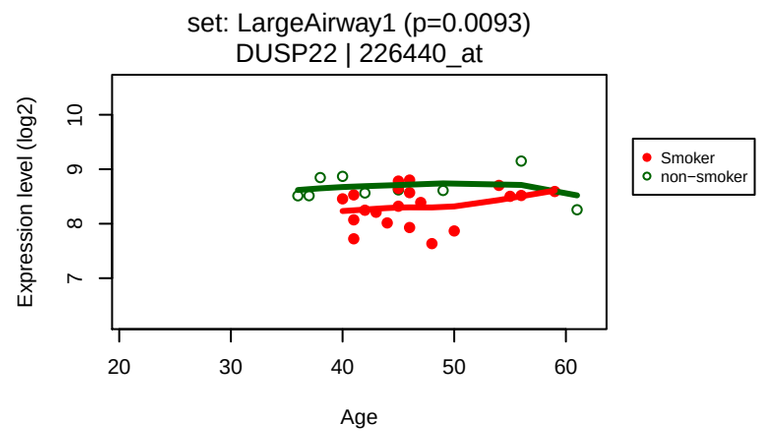

Gene expression of smokers and nonsmokers
gene: DUSP22 | 226440_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0397
p=0.0132
p=0.0073
p=0.0093
p=0.0667
p=0.0036
p=0.4458
p=0.2798
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

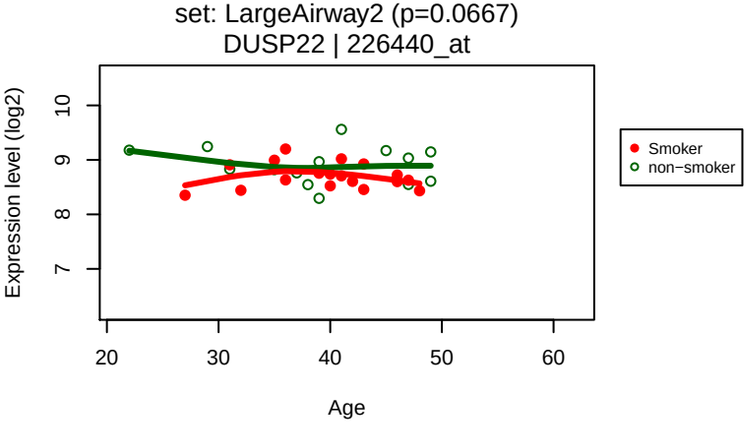

Gene expression ratio (smokers/nonsmokers)
gene: DUSP22 | 226440_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0397
p=0.0132
p=0.0073
p=0.0093
p=0.0667
p=0.0036
p=0.4458
p=0.2798
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

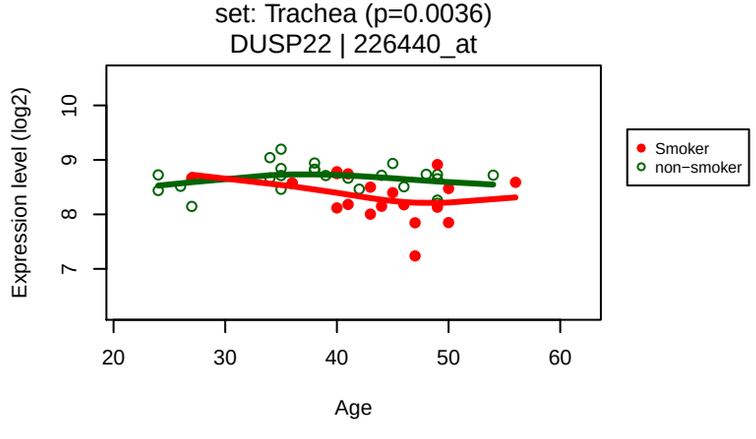

Gene expression ratio in Males
gene: DUSP22 | 226440_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0045
p=0.4564
p=0.0066
p=0.2786
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DUSP22 | 226440_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7922
p=1e−04
p=0.3518
p=0.8555
SmallAirway2
LargeAirway2
Trachea
Macrophage2

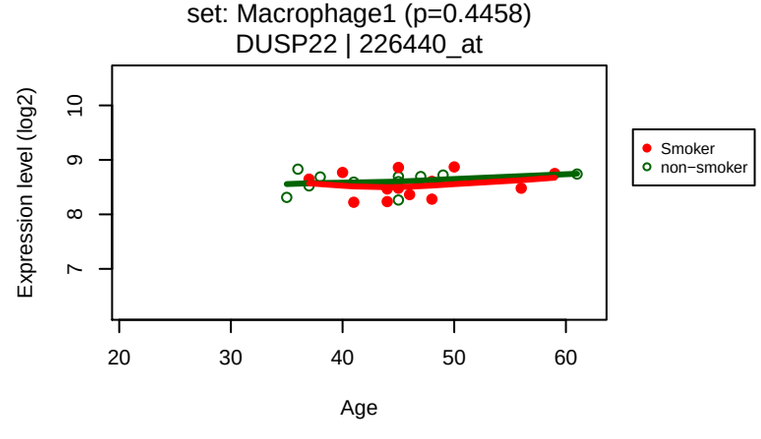

set: SmallAirway1 (p=0.0397)
DUSP22 | 226440_at
Expression level (log2)
Age
Smoker
non−smoker

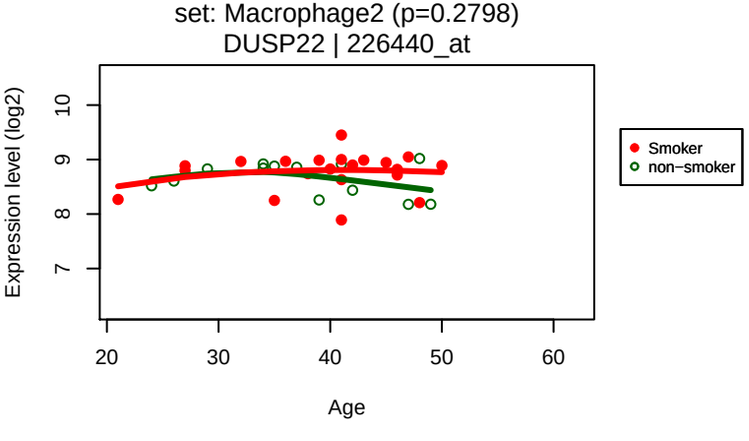

set: SmallAirway2 (p=0.0132)
DUSP22 | 226440_at
Expression level (log2)
Age
Smoker
non−smoker

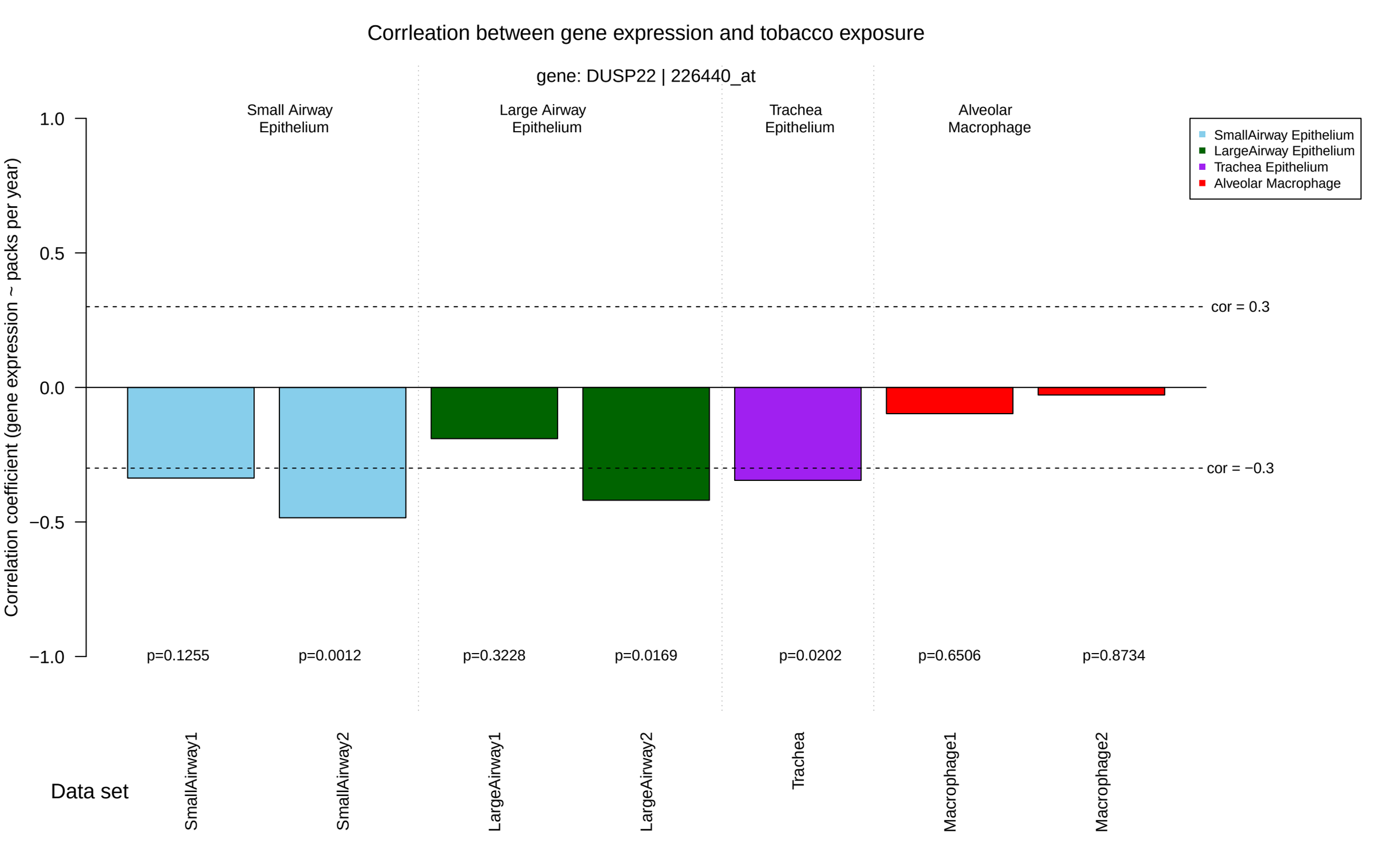

set: LargeAirway1 (p=0.0093)
DUSP22 | 226440_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0667)
DUSP22 | 226440_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0036)
DUSP22 | 226440_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4458)
DUSP22 | 226440_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2798)
DUSP22 | 226440_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DUSP22 | 226440_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1255
p=0.0012
p=0.3228
p=0.0169
p=0.0202
p=0.6506
p=0.8734
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2